

Dongfang Zhao, Ph.D.

Status: U.S. citizen, married with two children

Position: Assistant Professor of Computer Science, University of Nevada, Reno

Professional Preparation

Northeastern University	China	Computer Science	BE, 2003
Katholieke Universiteit Leuven	Belgium	Statistics	MS, 2005
Emory University	Atlanta, GA	Computer Science	MS, 2008
Illinois Institute of Technology	Chicago, IL	Computer Science	PhD, 2015
Pacific Northwest National Laboratory	Seattle, WA	Applied Cryptography	PostDoc, 2016
University of Washington	Seattle, WA	Distributed Databases	PostDoc, 2017

Employment

2017–2023	Assistant Professor of Computer Science, University of Nevada, Reno
2018–2020	Visiting Professor of Computer Science, University of California, Davis
2015	Research Intern, IBM Almaden Research Center, San Jose, CA
2014	Research Intern, Argonne National Laboratory, Chicago, IL
2013	Research Intern, Pacific Northwest National Laboratory, Seattle, WA
2009–2011	Software Developer, Epic Systems Corporation, Madison, WI
2006–2007	Research Assistant, Centers for Disease Control and Prevention (CDC), Atlanta, GA
2003–2004	Research Assistant, Interuniversity Microelectronics Centre (IMEC), Belgium

Awards and Grants

2022	Federal Research Grant, NSF 2112345, \$255,916, PI at University of Nevada “A DLT Machine Learning Platform for Blockchain Warehousing”
2020	Federal Research Grant, DOE DE-SC0020455, \$200,000, PI at University of Nevada “HPChain: Energy-Efficient Scalable Blockchains for HPC Systems”
2019	Best Paper Award, International Conference on Cloud Computing “BlockLite: Toward Accurate and Efficient Emulation of Public Blockchains in the Cloud”
2019	Industrial Research Grant, Amazon Web Services, \$30,000
2019	Industrial Research Grant, Google Cloud Platform, \$15,000
2018	Best Student Paper Award, IEEE International Conference on Cloud Computing “Toward Cost-Effective Memory Scaling in Clouds: Symbiosis of Virtual and Physical Memory”
2017	Industrial Research Grant, Microsoft Azure, \$20,000
2017	Industrial Research Grant, Amazon Web Services, \$10,000
2016	Postdoctoral Fellowship, Sloan Foundation, \$155,000
2007	Graduate Fellowship, Oak Ridge Institute for Science and Education, \$85,000

Teaching

CS 705: Cryptography and Blockchain	University of Nevada, Reno	2022–2023
CS 457: Database Management Systems	University of Nevada, Reno	2018–2023
CS 791: Distributed Computing	University of Nevada, Reno	2020–2022

Services

2021–2023	Editorial Review Board	IEEE Transactions on Distributed & Parallel Systems
2023	Program Committee	AAAI-23
2022	Grant Review Panel	National Science Foundation: CISE OAC
2022	Grant Review Panel	National Science Foundation: CISE IIS
2019	Organizing Committee	ACM/IFIP International Middleware Conference
2018	Grant Review Panel	U.S. Department of Energy: ASCR
2018	Grant Review Panel	National Science Foundation: CISE IIS

Selected Publications

Book chapters

- [1] **Dongfang Zhao**: Cross-Blockchain Transactions: Systems, Protocols, and Topological Theory. Handbook on Blockchain, Springer Optimization and its Applications. 2022.
- [2] **Dongfang Zhao**, Akash Mahakode, Sandip Lakshminarasiah, Ioan Raicu: High-Performance Storage Support for Scientific Big Data Applications on the Cloud. Resource Management for Big Data Platforms 2016: 147-170

Journals

- [1] **Dongfang Zhao**, Mohamed Mohamed, Heiko Ludwig: Locality-Aware Scheduling for Containers in Cloud Computing. IEEE Transactions on Cloud Computing. 8(2): 635-646 (2020)
- [2] **Dongfang Zhao**, Kan Qiao, Zhou Zhou, Tonglin Li, Zhihan Lu, Xiaohua Xu: Toward Efficient and Flexible Metadata Indexing of Big Data Systems. IEEE Transactions on Big Data 3(1): 107-117 (2017)
- [3] **Dongfang Zhao**, Ning Liu, Dries Kimpe, Robert B. Ross, Xian-He Sun, Ioan Raicu: Towards Exploring Data-Intensive Scientific Applications at Extreme Scales through Systems and Simulations. IEEE Transactions on Parallel and Distributed Systems. 27(6): 1824-1837 (2016)
- [4] **Dongfang Zhao**, Kan Qiao, Jian Yin, Ioan Raicu: Dynamic Virtual Chunks: On Supporting Efficient Accesses to Compressed Scientific Data. IEEE Transactions on Service Computing 9(1): 96-109 (2016)
- [5] **Dongfang Zhao**, Li Yang: Incremental Isometric Embedding of High-Dimensional Data Using Connected Neighborhood Graphs. IEEE Transactions on Pattern Analysis and Machine Intelligence. 31(1): 86-98 (2009)

Conferences

([†]Student advised by **Dongfang Zhao**)

- [1] [†]Abdullah Al-Mamun, Haoting Shen, **Dongfang Zhao**. DEAN: A Lightweight and Resource-efficient Blockchain Protocol for Reliable Edge Computing. The 36th IEEE International Symposium on Parallel and Distributed Processing (IPDPS), 2022.
- [2] [†]Olamide Timothy Tawose, [†]Bin Li, Lei Yang, Feng Yan, **Dongfang Zhao**. Topological Modeling and Parallelization of Multidimensional Data on Microelectrode Arrays. The 36th IEEE International Symposium on Parallel and Distributed Processing (IPDPS), 2022. Direct acceptance without revision
- [3] [†]Abdullah Al-Mamun, Feng Yan, **Dongfang Zhao**: SciChain: Blockchain-enabled Lightweight and Efficient Data Provenance for Reproducible Scientific Computing. 37th IEEE International Conference on Data Engineering (ICDE), 2021.
- [4] [†]Abdullah Al-Mamun, Feng Yan, **Dongfang Zhao**: BAASH: lightweight, efficient, and reliable blockchain-as-a-service for HPC systems. The International Conference for High Performance Computing, Networking, Storage and Analysis (SC), 2021. Direct acceptance without revision
- [5] [†]Xinying Wang, [†]Olamide Timothy Tawose, Feng Yan, **Dongfang Zhao**: HDK: Toward High-Performance Deep-Learning-Based Kirchhoff Analysis. The Thirty-Fourth AAAI Conference on Artificial Intelligence (AAAI), 2020. Oral presentation
- [6] [†]Abdullah Al-Mamun, Jun Dai, Xiaohua Xu, Mohammad Sadoghi, Haoting Shen, and **Dongfang Zhao**. DEAN: A Blockchain-Inspired Consensus Protocol Enabling Trustworthy Edge Computing. Network and Distributed System Security Symposium (NDSS). 2020.
- [7] Haoting Shen, Shahriar Badsha, and **Dongfang Zhao**. Poster: Consortium Blockchain for the Assurance of Supply Chain Security. Network and Distributed System Security Symposium (NDSS). 2020.
- [8] [†]Xinying Wang, [†]Abdullah Al-Mamun, Feng Yan, **Dongfang Zhao**: Toward Accurate and Efficient Emulation of Public Blockchains in the Cloud. The 12th International Conference on Cloud Computing (CLOUD), 2019. Best paper award
- [9] Heyang Qin, Syed Zawad, Yanqi Zhou, Lei Yang, **Dongfang Zhao**, Feng Yan: Swift machine learning model serving scheduling: a region based reinforcement learning approach. The International Conf. for High Performance Computing, Networking, Storage and Analysis (SC), 2019.
- [10] [†]Xinying Wang, Cong Xu, Ke Wang, Feng Yan, **Dongfang Zhao**: Toward Cost-Effective Memory Scaling in Clouds: Symbiosis of Virtual and Physical Memory. The 11th IEEE International Conference on Cloud Computing (IEEE CLOUD), 2018. Best student paper award
- [11] Parmita Mehta, Sven Dorkenwald, **Dongfang Zhao**, Tomer Kaftan, Alvin Cheung, Magdalena Balazinska, Ariel Rokem, Andrew J. Connolly, Jacob VanderPlas, Yusra AlSayyad: Comparative Evaluation of Big-Data Systems on Scientific Image Analytics Workloads. Proceedings of the VLDB Endowment (VLDB). 10(11): 1226-1237 (2017)
- [12] Tonglin Li, Xiaobing Zhou, Kevin Brandstatter, **Dongfang Zhao**, Ke Wang, Anupam Rajendran, Zhao Zhang, Ioan Raicu: ZHT: A Light-Weight Reliable Persistent Dynamic Scalable Zero-Hop Distributed Hash Table. The 27th IEEE International Symposium on Parallel and Distributed Processing (IPDPS), 2013.